

GO Developer Technical Assignment

Objective: Build a high-performance dashboard to analyze and visualize key business metrics from a provided dataset.

Business Context

ABT Corporation has provided a large dataset containing information about products, transactions, and customer regions. The company requires a data analytics dashboard that delivers clear and actionable insights based on this data.

Field	Description
transaction_id	Unique transaction ID
transaction_date	Date of purchase
user_id	ID of user making the transaction
country	Country of purchase
region	Region/state of purchase
product_id	Product ID
product_name	Product name
category	Product category
price	Region-adjusted price
quantity	Quantity purchased
total_price	price × quantity
stock_quantity	Original product stock
added_date	Product added date

Insights to be Displayed

Your dashboard should include the following components:

1. **Country-Level Revenue Table**

A table that lists:

- **Country**
- **Product Name**
- **Total Revenue**
- **Number of Transactions**

2. This table must be **sorted by total revenue in descending order**, with the highest-revenue countries at the top.

3. **Top 20 Frequently Purchased Products**

Display the **top 20 most frequently purchased products** along with their **current available stock quantity** using an appropriate and informative chart (e.g., bar chart, horizontal bars, etc.).

4. Develop a chart to highlight the months with the highest sales volume.

5. **Top 30 Regions by Revenue**

Similarly, visualize the **top 30 regions** with the highest total revenue and items sold using an appropriate chart.

Technical Requirements

- **Performance**
 - The dashboard must **load within 10 seconds**.
- **Reusability & Extensibility**
 - The solution must be **reusable**. ABT may request additional insights in the future, so the codebase and logic should support easy extension.
- **Documentation & Delivery**
 - Submit **all files used** in the development of your solution.
 - If any **data manipulation or preprocessing scripts** are used, include these scripts in your submission with clear documentation.
- **Flexibility in Data Handling**
 - You are free to perform **data transformation, optimization** as needed to meet the requirements. Please ensure all such steps are documented.
- **Testing**
 - Write Unit testing & make a test coverage report.

Technology Notes

- You may use **Go for backend logic and data processing**.
- You can choose any visualization framework for the frontend.

Submission

Please submit the following:

- Source code and scripts
- README or setup guide
- Any additional assets used